

FT Player — Oh! This is the Fourier Transform!

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ABSTRACT

The Fourier transform process is difficult to represent in the two-dimensional plane, and beginners have difficulty understanding the concept of transformation from the time domain to the frequency domain. In this paper, we propose a Virtual Reality instructional tool designed to facilitate a better understanding of the mathematical concepts and processes involved in the Fourier transform. To illustrate the winding process, users can interact with the controller to adjust the winding frequency. Through pushing and pulling actions, users can synthesize and decompose waves, visually represented by graphs of winding centroids. The virtual environment offers intuitive demonstrations and visualization tools that demystify the abstraction associated with the Fourier transform, enabling users to comprehend the underlying mathematical principles more effectively.

Index Terms: Fourier Transform; Virtual Reality; Mathematics Education

1 INTRODUCTION

To address the difficulties in elucidating the Fourier transform with traditional teaching methods and offer a more intuitive and immersive learning experience, we have designed and developed a virtual reality-based instructional tool for the Fourier transform. The learning process is structured into three stages. In Stage 1, users superimpose waves by adding several sine waves through the panel, forming a combined wave. Stage 2 involves the transformation of waves, where users wrap the combined wave around a circle by adjusting the winding frequency. In Stage 3, users engage in wave decomposition by pushing and pulling to decompose the sine and frequency components of the combined wave.

Throughout this learning process, users can experience the transformation between the time and frequency domains, observing changes in the signal spectrum and gaining a profound understanding of the mathematical principles underlying the Fourier transform.

2 RELATED WORK

Del Cerro Velázquez, F. et al. proposed Geogebra AR, an application centered around mobile devices [1]. The program promotes the learning and analysis of mathematical functions. It can help students visualize how a change in one variable immediately affects other variables, improving their learning experience and understanding of functions more than solving formulas to get answers. However, educators need some training to better integrate the program into the teaching and learning process.

3 CONCEPT

3.1 Hardware and Software

The solution uses the Oculus Quest 3 as the virtual reality device and the system was developed using Unity 2021.3.33 and Oculus Integration SDK 57.0.

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3.2 The Fourier Transform

3.2.1 Classic Fourier Transform

The Fourier transform is a linear integral transform used to transform a function between the time and frequency domains. Its mathematical formula is: $f(\xi) = \int_{-\infty}^{\infty} g(x) e^{-2\pi i x \xi} dx$.

The Fourier transform derives from the study of the Fourier series. In the Fourier series, complex periodic functions can be represented as a series of simple sums of sine or cosine waves. The Fourier transform is an extension of the Fourier series in that the period of the function represented by it tends to infinity.

3.2.2 Almost Fourier Transform

The Almost Fourier Transform is a mathematical transform that approximates the Fourier Transform. To better explain the Fourier Transform using a graphical representation, we introduce the Almost Fourier Transform, whose mathematical formula is: $\tilde{g}(f) = \frac{1}{t_1 - t_2} \int_{t_1}^{t_2} g(t) e^{-2\pi i f t} dt$.

It corresponds physically to the center of mass of the function expression in the mathematical formula: $g(t) e^{-2\pi i f t}$. The main difference between the almost Fourier transform and the Fourier transform is the range of the integral: the former integrates over a finite time range, while the latter integrates over an infinite domain of definition. In addition, the mathematical formula for the almost-fourier integral: $\frac{1}{t_1 - t_2}$ partially decreases the value of its function as the range of the time domain increases.

3.2.3 Winding Frequency and Initial Frequency

For ease of presentation and understanding, the original functions used in the program are all periodic functions. The initial frequency refers to the frequency of the original periodic function: $g(t)$.

"Winding" is a graphic description of the process of changing the image of the expression mathematical formula: $g(t) e^{-2\pi i f t}$ in the complex plane as the frequency changes. The winding frequency is the frequency of the winding process, i.e., the parameter in the complex exponential term mathematical formula: $e^{-2\pi i f t}$ mathematical formula: f .

3.3 Design

We organize and illustrate the projection relationship between the stages of the Fourier transform and the Fourier transform process in three-dimensional space. The framework is primarily segmented into three modules: the input module, the visualization module, and the auxiliary display module, as illustrated in Figure 1.

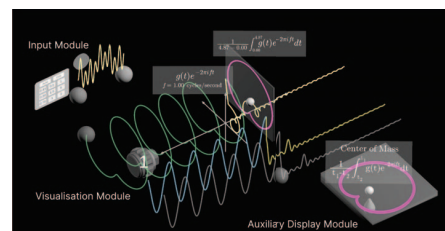


Figure 1: 3D user interface and components of FT Player

3.3.1 The Input Model

The input module contains a sinusoidal function-adding panel where the user can define the function's amplitude, frequency, and phase. Positioned behind the left side of the coordinate axis, the combined wave module generates a superimposed waveform based on the user's input.

3.3.2 The Visualization Module

The visualization module comprises two parts: wave transformation and wave decomposition. In the wave transformation segment, three-dimensional winding graphs of waves are displayed on the left side of the coordinate axis, with accompanying wave graphs following the three-dimensional winding patterns. These three-dimensional winding graphs are then projected onto the right side of the coordinate axis, forming two-dimensional winding graphs. The position of the center of mass of these two-dimensional winding graphs changes with the frequency, creating an image of winding centroids. The graphs of winding centroids are further projected to the rear as projection graphs.

Users can rotate the control, altering the winding frequency of the combined wave, and shift the control to modify the function's time-domain range. This section illustrates the process of changing the winding graphs of waves and winding centroids. In the wave decomposition segment, through interactive pushing and pulling actions, users can observe the sine and frequency components of the combined wave. This showcases the Fourier transform conversion process from the time domain to the frequency domain.

3.3.3 The Auxiliary Display Module

The auxiliary display module contains the centroid identifier, which uses the gravity balance support point of the plate to represent the relationship between the centroid and the image. The image on the plate is the image formed by the function winding around a circle, and the position of the fulcrum represents the centroid of the image.

4 INTERACTION

The system's interaction is facilitated through the controller. We have devised three interaction modes: winding, pushing, pulling, and traversing. These modes correspond to wave winding, wave synthesis and decomposition, and the transformation of the wave's time interval during the transformation process.

4.1 Winding

As depicted in Figure 2, the user uses the trigger key to grab the wind control. Turning clockwise increases the twisting frequency, and turning counterclockwise decreases it. (Release diagram) The twist helps the user control the winding frequency more accurately. Relating the interaction to the winding of a wave facilitates the user's understanding of the wave-winding process.

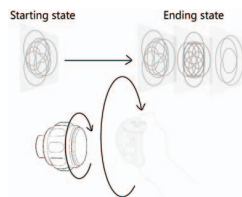


Figure 2: Clockwise winding to increase the twisting frequency

4.2 Pushing and Pulling

The push and pull of the drawer closes and opens the drawer. The user can push and pull to complete the synthesis and decomposition of the wave and understand the Fourier transform and Fourier inverse

transform processes, as depicted in Figure 3. During the transformation process, the user uses the trigger key to grab the push-pull control and pull it out in the axial direction (put the schematic). The combined wave is Fourier-transformed to obtain multiple decomposed waves. The reverse is done to obtain a combined wave. This operation is easy to understand, user-friendly, and interesting.

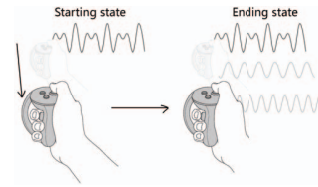


Figure 3: Pulling to decompose the combined wave

4.3 Traversing

The time domain range is an important factor affecting the classic Fourier transform. The user can change the time domain range of the original function by using the trigger key to grab the position of the traverse control, as shown in Figure 4. When the parameter is changed, the value in the formula displayed above is also highlighted and changed.

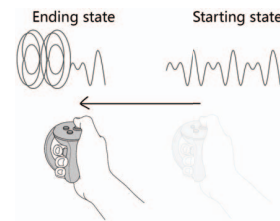


Figure 4: Traversing to change the time-domain range of the wave

5 INNOVATION

5.1 Intuitive Correspondence in the Time and Frequency Domains

The educational tool illustrates the correlation between time and frequency domain graphs in three-dimensional space. Users can dynamically observe the conversion process between the frequency domain and the time domain by adjusting both the winding frequency parameter and the signal's time parameter. This presentation method effectively decomposes the learning barrier for students faced with the complexity of the Fourier transform, aiding them in achieving a deeper understanding of the essence of the Fourier transform.

5.2 Physical Metaphor Interaction Design

The teaching tool integrates metaphors from real-life physical laws of motion to establish an intuitive and natural interaction environment for users, where controls move in alignment with the manipulated object. Parameters are adjusted coherently, facilitating easy and swift interaction with the object through the natural manipulation of the controls.

REFERENCES

- [1] F. del Cerro Velázquez and G. Morales Méndez. Application in augmented reality for learning mathematical functions: A study for the development of spatial intelligence in secondary education students. *Mathematics*, 9(4), 2021.